(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: ES-201 **Subject: Computational Methods**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data if any. Scientific calculator is allowed.

Q1 Attempt all questions:

a) The height of an observation tower was estimated to be 47m, whereas its actual height was 45m. Calculate the percentage relative error in the measurement. **(1.5)**

b) Derive the formula for evaluating $\sqrt{12}$ by Newton's Raphson method. **(1.5)**

c) Differentiate between partial and complete pivoting in solving linear system of algebraic equations. **(1.5)**

d) Estimate the missing term in the following table: **(1.5)**

x:	0	1	2	3	4
$f(x)$:	1	3	9	__	81

e) Define cubic spline function and state the conditions required for cubic spline interpolation. **(1.5)**

f) Find $\int_0^{\pi/2} \sqrt{\cos\theta}\, d\theta$ using Simpson's 1/3rd rule for $n=6$. **(2.5)**

g) Solve the IVP $\frac{dy}{dx} = xe^y, y(0)=0$ using Picard's method and estimate $y(0.2)$. **(3)**

h) Determine which of the following equations are elliptic, parabolic, and hyperbolic. **(2)**

(i) $f_{xx}+6f_{xy}+9f_{yy}=0$

(ii) $f_{xx}-2f_{xy}+2f_{yy}=2x+5y$

UNIT-I

Q2 a) Find the root of the equation $x=e^{-x}$, correct to three decimal places by Secant method by performing six iterations. **(8)**

b) Use Newton—Rapshon method to obtain a root of $\sin x = 1-x$ to three decimal places. **(7)**

Q3 a) Perform four iterations of Golden section search method to minimize $f(x)=x^4-14x^3+60x^2-70x$, $x\in(0,2)$ with $\varepsilon=10^{-3}$. **(7.5)**

b) Use steepest descent method for 3 iterations on $f(x_1,x_2,x_3)=(x_1-4)^4+(x_2-3)^2+4(x_3+5)^4$ with initial point $x^{(0)}=[4,2,-1]^T$ **(7.5)**

UNIT-II

Q4 a) Prove that $\Delta=\frac{1}{2}\delta^2+\delta\sqrt{(1+\delta^2/4)}$. **(7)**

b) The table gives the distance in nautical miles of the visible horizon for the given heights in feet above the earth's surface: **(8)**

P.T.O.